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Continuous Assessment 3 (CA 3) - Questions

Course Code: CSE563 | Course Title: APPLIED MATHEMATICS

Section(s): K3P25RB | Syllabus Covered: Unit 5 and Unit 6

Total Marks: 30 Time Allowed: 60 minutes | Maps to CO5 and CO6 - algorithmic complexity; optimization approaches

Unit V: Algorithm Complexity Analysis · Unit VI: Algorithmic and Optimization Approach

Instructions: attempt all questions. Show all intermediate recurrence derivations and DP tables clearly. Unsupported answers receive no credit. The marks for each question are shown in brackets.

Part A - Short Answer Questions (6 Marks - 3 questions x 2 marks)

1. For each recurrence, state a, b and f(n), identify the Master theorem case, and give the asymptotic bound: (a) $T(n) = 8T(n/2) + n^2$; (b) $T(n) = 4T(n/2) + n^2$. **(2 Marks)**

2. A fractional knapsack has capacity W = 25 kg. Three items are available: Item 1 (10 kg, $60); Item 2 (15 kg, $75); Item 3 (20 kg, $80). Compute the value-to-weight ratios, state the greedy order, and calculate the maximum profit. **(2 Marks)**

3. (a) State the worst-case time complexity of merge sort and quick sort, and the condition causing quick sort's worst-case degradation. (b) State Karatsuba's integer multiplication recurrence and its asymptotic bound. **(2 Marks)**

Part B - Core Numerical Problems (12 Marks - 2 questions x 6 marks)

4. Recurrence relations and formal asymptotic analysis: **(6 Marks)**

(a) Solve $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, with $a_0 = 1$ and $a_1 = 4$. Write the characteristic equation, find the roots, derive the closed form, and verify it for a_2 and a_3.

(b) Use the formal definition of big-Θ to prove that $3n^2 + 5n + 7 = \Theta(n^2)$. Explicitly find constants $c_1, c_2 > 0$ and a threshold n_0 such that $c_1n^2 \leq 3n^2 + 5n + 7 \leq c_2n^2$ for all $n \geq n_0$.

5. Dynamic programming for the 0/1 knapsack. Capacity W = 6, with four items: Item 1 (w = 1, v = 10), Item 2 (w = 2, v = 15), Item 3 (w = 3, v = 40), Item 4 (w = 4, v = 50). **(6 Marks)**

(a) State the DP recurrence relation for the table entry V[i, w].

(b) Construct and completely fill the dynamic programming table for all $0 \leq i \leq 4$ and $0 \leq w \leq 6$.

(c) Perform a traceback from V[4, 6] to find the optimal subset of selected items, the total weight, and the maximum total profit.

Part C - Applied and Analytical Problems (12 Marks - 2 questions x 6 marks)

6. Boolean satisfiability and computational complexity theory: **(6 Marks)**

(a) Let $\varphi(x_1, x_2, x_3) = (x_1 \vee x_2 \vee x_3) \wedge (\neg x_1 \vee \neg x_2) \wedge (\neg x_2 \vee \neg x_3) \wedge (x_1 \vee \neg x_3)$. (i) Evaluate whether $(x_1, x_2, x_3) = (T, F, T)$ satisfies φ, showing each clause. (ii) Conclude whether φ is satisfiable.

(b) Define the complexity classes P, NP and NP-complete. (i) Distinguish between polynomial-time verification and solvability. (ii) If an $O(n^3)$ algorithm solves 3-SAT, explain what this implies for P against NP, using polynomial reductions.

7. Nature-inspired optimisation and genetic algorithms: **(6 Marks)**

(a) A GA maximises $f(x) = x^2$ over $x \in [0, 31]$ with 5-bit chromosomes: $C_1 = 01100$ (x = 12), $C_2 = 11001$ (x = 25), $C_3 = 00111$ (x = 7), $C_4 = 10010$ (x = 18). (i) Compute the fitness of each and the roulette-wheel selection probabilities. (ii) Crossing C_2 (110|01) with C_4 (100|10) after bit 3, write the offspring, evaluate their fitness, and show that crossover produced an offspring superior to both parents.

(b) Simulated annealing with Metropolis acceptance for $\Delta E = +15$: (i) Calculate P(accept) at T = 100. (ii) Calculate P(accept) at T = 5. (iii) Explain how cooling balances exploration against exploitation and avoids local-optimum traps.

----- End of Question Paper -----